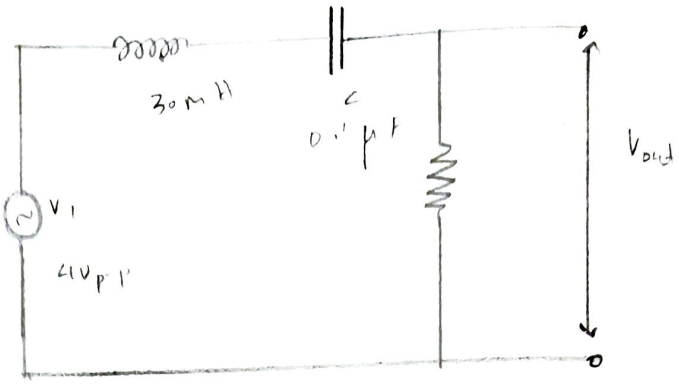
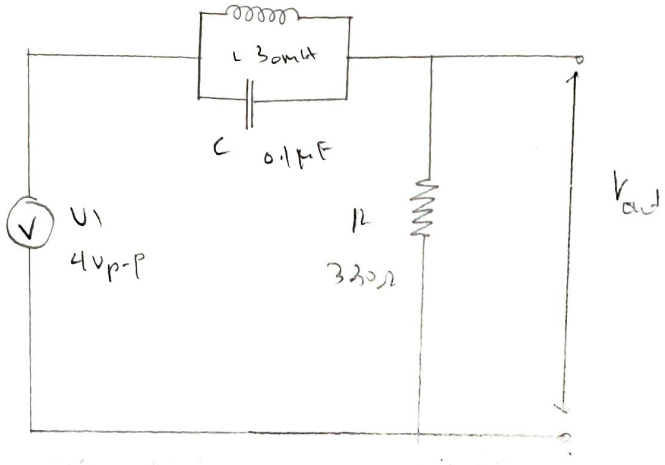
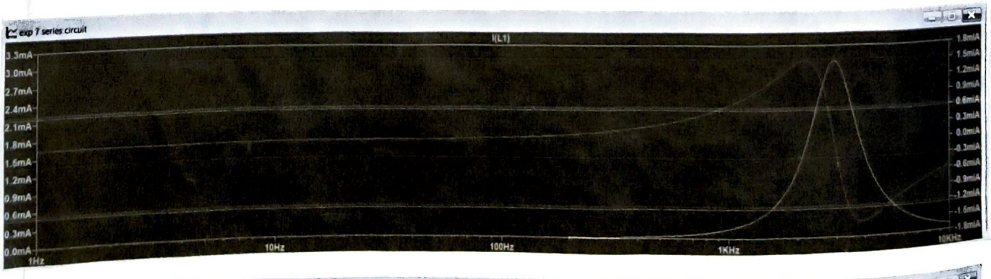
circuit Diagram:



series RLC circuit



Parallel RLC Circuit



Exp No: 7
Date: 28/02/2026

Frequency Responce of Series And Parallel resonance circuits.

Aim:

* To study the behaviour of series and parallel RLC circuit at resonance and to determine the resonant frequency, bandwidth and a of the given RLC circuit using multisim simulator.

Apparatus Required:

S.No	Components	Specification	Quantity
1	CRO	10 MHz	1
2	DIB	–	1
3	capacitor	0.1 μF	1
4	Resistors	330 Ω	1
5.	Bread Board	–	–
6.	connections wires	–	
7.	PC with multisim simulator	–	

Formula:

$$d_0 = \frac{1}{2\pi\sqrt{LC}}$$

$$\text{Bandwidth} = d_2 - d_1$$

$$Q = d_0 / BW$$

Tabulation:

Series Resonance		Parallel Resonance	
Frequency (Hz)	output current (mA)	Frequency (Hz)	output current (mA)
500Hz	0.04 mA	500Hz	2.9mA
1KHz	0.16 mA	1KHz	2.54 mA
1.5KHz	0.46mA	1.5KHz	1.96mA
2KHz	1.15mA	2KHz	1.26mA
2.9KHz	8mA	2.9KHz	0.002 mA
3.5KHz	2.18mA	3.5KHz	0.6mA

Calculation:

theoretical calculation:

for series

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

$$= \frac{1}{2\times\pi\sqrt{80\times10^{-3}\times0.1\times10^{-6}}}$$

$$f_r = 2.9\,KHz$$

$$\text{Band width} = \frac{R}{2\pi L}$$

$$= \frac{880}{2\pi \times 8.0\times10^{-3}}$$

$$B_w = 1.75\,KHz$$

Quality Factor:

$$Q = \frac{f_r}{B_w}$$

$$= \frac{2906}{1750}$$

$$Q = 1.66$$

Parallel circuit

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

$$f_r = 2.9\,KHz$$

Band width:

$$B_w = \frac{1}{2\pi RC}$$

$$= \frac{1}{2\times\pi\times330\times0.1\times10^{-6}}$$

$$B_w = 4.82\,KHz$$

Quarity Factor:

$$Q = \frac{f_r}{B_w} = \frac{2906}{482} = 0.6$$

S.No	Description	Marks Allotted	marks obtained
1.	Experimental setup	20	20
2.	observation	20	18
3.	calculation	20	18
4.	viva	20	18
5.	Record	20	20
			94

Result.

* thus the phenomenon of resonance in RLC circuit was studied and the following were determined using multisim simulator

Series RLC	Parallel RLC
Resonant frequency - 2.9KHz	Resonant frequency = 2.9KHz
Bandwith - 1.75KHz	Bandwith - 4.8242
Q-factor - 1.66	Q-factor - 0.6